THIS PCB's PURPOSE or DESIGN GOAL:

* update MITS/Altair CPU-A board for modern EMC PCB layout principles
  ** power plane(s) to reduce Vcc and Ground shifting niose
  ** place decopupling capacitors next their respective ICs
* improve processing speed above 1975 technology limited 8080A with newer CPU
  ** HMOS 8085A upto 6MHz (late 1970s)
  ** CMOS 80C85B upto 8MHz (mid 1990s)
* add low voltage or 5Vcc < 4.7V forced RESET to help power up reliability
* same PCB to support replacement of IMSAI 8080A MPU-A
  ** have 2 ribbon cable connectors
  ** S-100 jumpers and circuits for supporting signal varations
* add Power On Jump to a user defined address (1K increments)
* minimal hardware, no IO, no RAM, no ROM, etc... just CPU
* allow for easy changing of CPU MHz speed
  ** use DIP oscillator as flexible source
  ** have PCB holes to allow HC case quartz crystal
* design a spin off varation to use the NSC800 (a Z80A hybrid with 8085 like pin out)

THIS DOCUMENT'S CONTENTS:

Top Sheet, Repro update
Altair CPU-A with 8085A/80C85B

Sheet: /
File: CPU-85_r104.kicad_sch

Title:

Size: A4 | Date: 18 July 2026, greglori_mi@wowway.com | **Rev:** rev 1.04

KiCad E.D.A. 10.0.4 | Id: 1/12

CHANGE HISTORY:

6/13/2026 thru 7/4/2026 rev 1.0
* started to make from IMSAI MPU-B rev 3.43 schematic by deleting parts
* made new unique oscillator foot print to allow use of either 8 or 14 pin DIP package
* changed all but 1 twin 2 pin jumpers (like J2 and J3) to 3 pin jumper
* redesigned CPU state outputs to Front Panel based on MPU-A, CPU-A and MPU-B schematics
* added 8 pin Altair Front Panel connector to support both IMSAI and MITS/Altair applications
* added page 7 Power On Jump feature
* 4 layer Gerbers made, no routing or DRC issues
* no PCBs made

7/4/2026 rev 1.01
* changed PCB to a 2 layer PCB
* 2 layer Gerbers made, no routing or DRC issues
* no PCBs made

7/8/2026 rev 1.02
* added C26 for slower CPU crystals
* 2 layer Gerbers made, no routing or DRC issues
* changed 30u" (Military/Medical grade) to 20u" (Industrial grade) Hard Gold on S-100 fingers
* schematic presented to S-100 Vintage Computer Group for review and comments
* no PCBs made

7/9/2026 to 7/13/2026 rev 1.03
* add CPU State Latch for driving CPU status sM1 on S-100 Bus DataOut5 for Front Panels
* made new PSYNC signal, 3 selectable sources: IMSAI MPU-B ALE as PSYNC made a PSYNC 25% shorter; or M1 from CPU State Decoder; or a pulse lengthened to the end of T2 cycle
* renumbered U23 thru U26, made space for added Tri State driver latch U23
* two 8T97 changed to 74HCT367, worked OK on existing IMSAI MPU-B at 4MHz
* newer KiCAD 10.99 was not compatible with FreeRouter 2.2.4, so had to re-load KiCAD 10.0.4
* deleted D5 and R55 POJ indicator LED
* FreeRoputer 2.2.4 not performing autorouting operation (still invenstigating on 7/13/2026)
* schematic given to S-100 Vintage Computer group for review and to identify issues
* no PCBs made

7/14/2026 to 8/8/2026 rev 1.04
* 8080A CPU State Table needs to be implemented not the 8085 CPU State Table in U23
* U23 changed from 74HCT138 to 16V8 that can perform some complex gating
* feedback sINTA is now a hardwired 16V8 input because WINCUPL did not let me use an 16V8 internal path
* restored D5 and R55 POJ indicator LED
* auto routing problem with FreeRouter versions between 2.1.0, 2.2.0 and 2.2.4 stalls with unknown problem... Free Router team made aware of the version sensivity and given the DNS files to help debug this is FreeRouter Bug Report #753
* success on 2 layer with Ground plane and 4 layer using older 2.1.0 Free Router
* 2 and 4 layer versions PCB layouts made
* only 2 layer Gerbers sent out for fabrication as 1.04.2
* andrasfuchs emailed issue #753 fixed in upcoming new FreeRouter release v2.3.x(?) (4Aug2026)

Reference Specifications:

Mechanical Outline is from IEEE 696 S-100 Bus Specification 1983-06_Images.pdf

NOTE1: Bottom S-100 Edge, 45 degree chamfer, 0.60mm into the top and bottom faces, 1/3 depth or 0.6mm of PCB thickness; total PCB thickness 1.60mm to 1.80mm or 0.066inch

NOTE2: 20u" Hard Gold on S-100 fingers with "lmm" finish

NOTE3: PCB material is FR4 - 170TG with >200V isolation, UL 94V-0 Flammability

NOTE4: 2oz copper front and back layers, inner layers 1oz
Green LPI masking, via holes plugged with masking
white silk screen text

NOTE5: IMSAI MPU-B Schematic Rev 2.pdf as updated by Robret Weatherford 3/2021

NOTE6: IMSAI MPU-B Reference Manual

NOTE7: page 24 880-101 CPU-A card schematic from MITS_Altair8800TheoryOperation_1975.pdf

NOTE8: page 34 of S-100 Bus Handbook by Dave Bursky, Hyaden Book Company, Rochelle Park, NJ

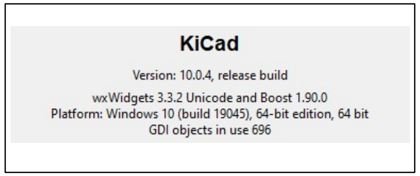
NOTE9: https://www.telewiretech.com/blogs/technical-resources/the-definitive-guide-to-connector-plating-thickness-15%CE%BC-gold-30%CE%BC-gold-and-tin-compared

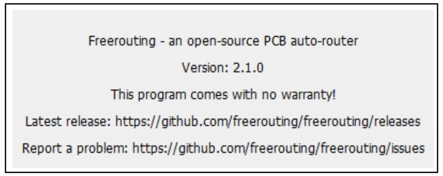
NOTE10: Cromemco ZUP (Z80A) card schematic

NOTE11: IMSAI Schematics, Page 4 is MPU-A rev4 from 2/76

NOTE12: Intel MCS80 8080 MIcrocontroller Systems User's Manual September 1975

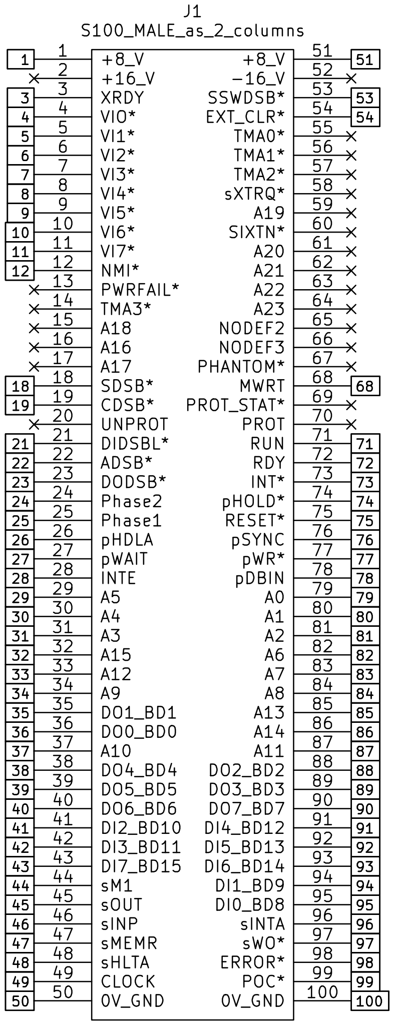
SOFTWARE VERSIONS USED:





S-100 PCB Card Information

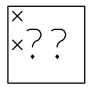
Sheet: /Change_History_and_Specs/		
File: S100_Card_sch.kicad_sch		
Title:		
Size: A4	Date: 4 Aug 2026, greglori_mi@wowway.com	**Rev:** rev 1.04
KiCad E.D.A. 10.0.4		Id: 2/12



NOTE: S–100 Labels and PCB Holes Cut & Pasted from
John Monohan's web page https://avitech.com.au/?page_id=3240
(Google Vintage Computer Group, used on IMSAI CPU–A repro panel)
26April2025

That symbol re–imported into KiCad, and the symbol editted from 1 column of 100 to 2 columns of 50 format to allow fitting on A sized page; DI, DO and 16 bit pins renamed per page 18 of IEEE–696 standard; a few pin names corrected ei. IEEE called pins 92 and 93 BiDirectional 3 "even" and BiDirectional 4 "odd"... which looks to be typo errors to me.
22June2026

Card ejector mounting holes

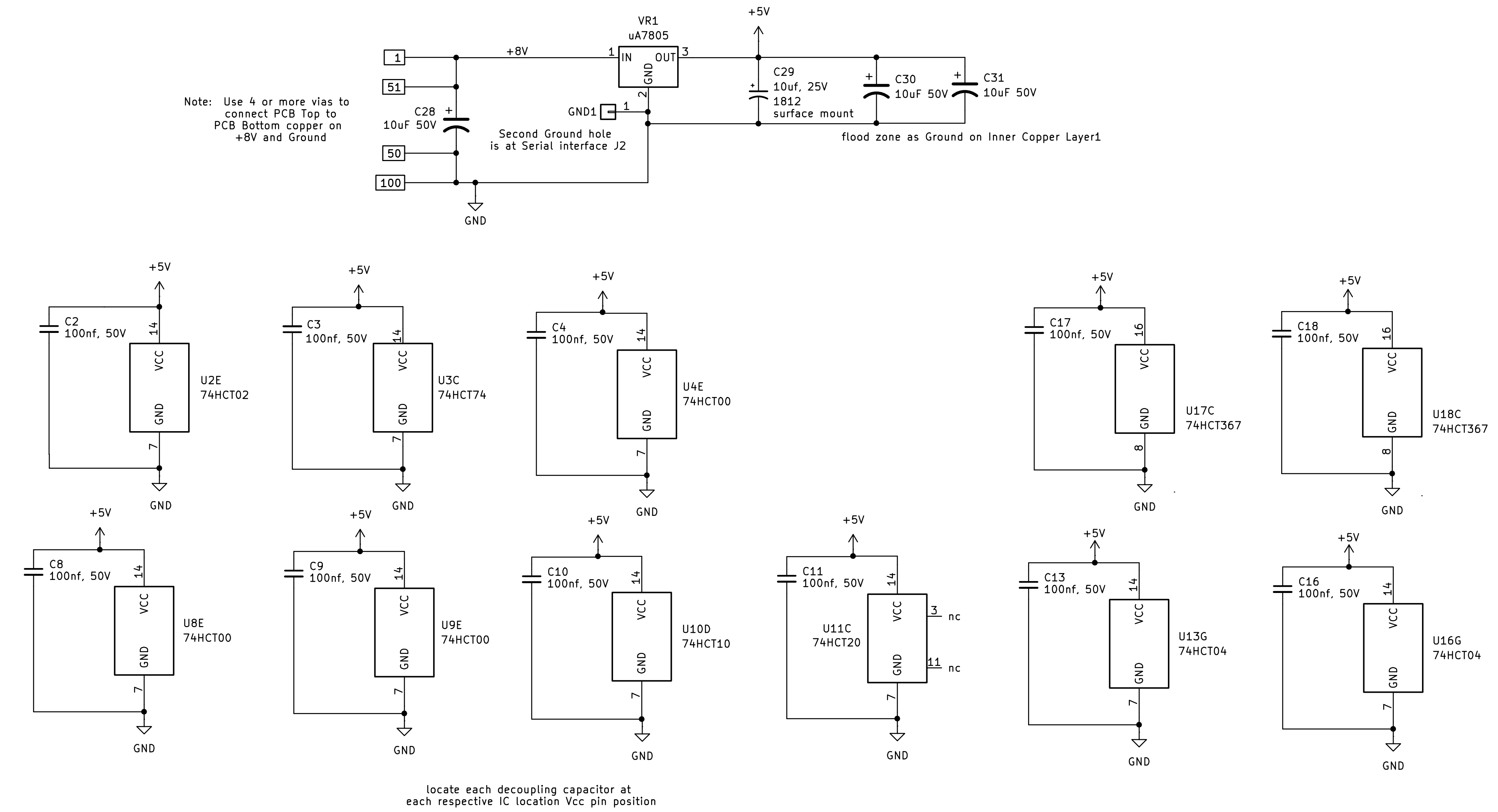


H1andH2

NOTE: S–100 pin number labels are used to connect to circuits on these schematic pages.

S–100 Connector Assignments

Sheet: /S–100_Bus_Connector/		
File: S100_Bus.kicad_sch		
Title:		
Size: A4	Date: 17 July 2026, greglori_mi@wowway.com	**Rev:** rev 1.04
KiCad E.D.A. 10.0.4		Id: 3/12

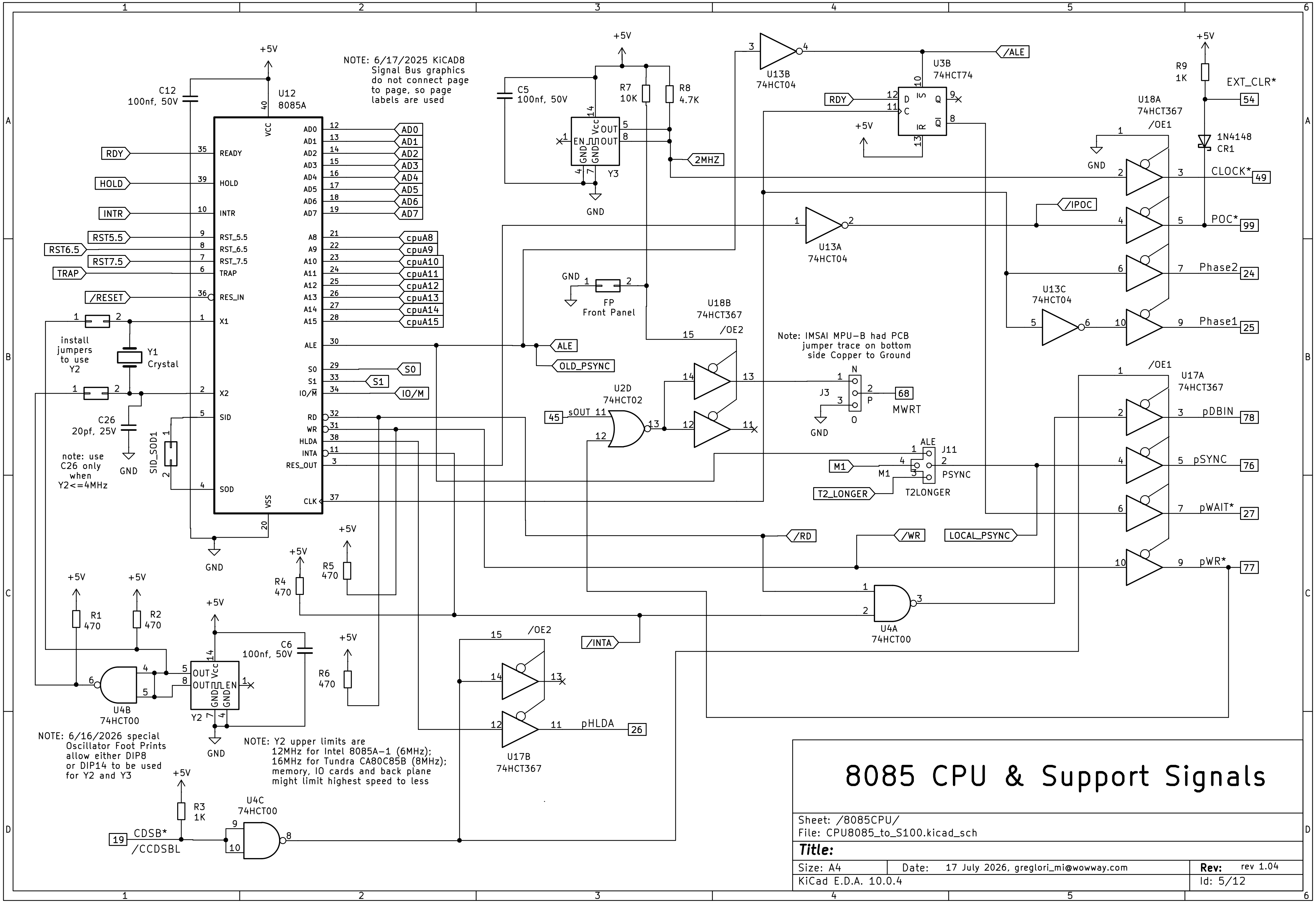


5V Regulator & Decoupling Caps

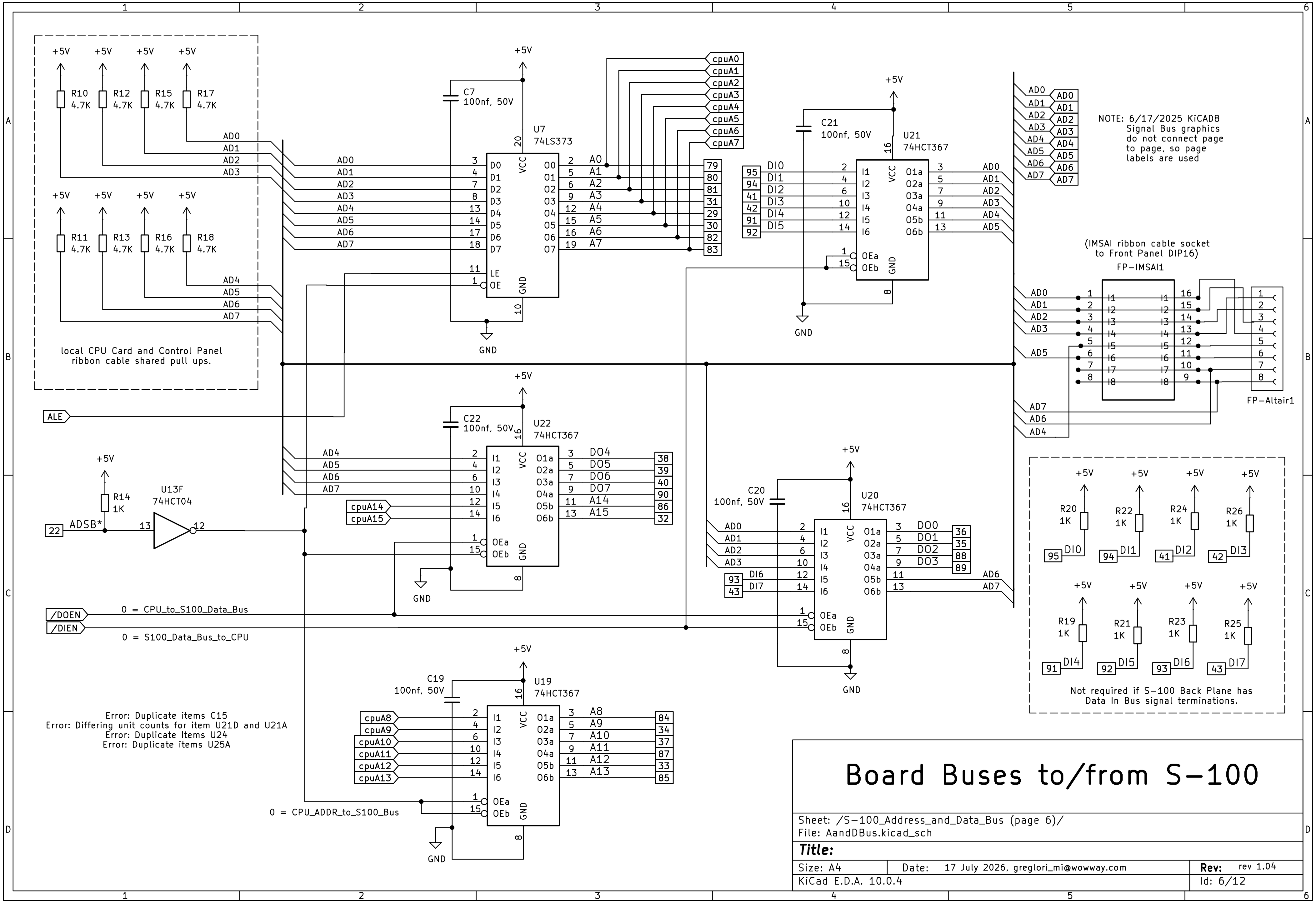
Sheet: /Power_Supplies/
File: Power_supplies.kicad_sch

Title:

Size: A4	Date: 17 July 2026, greglori_mi@wowway.com	Rev: rev 1.04
KiCad E.D.A. 10.0.4		Id: 4/12



NOTE: 6/17/2025 KiCAD8 Signal Bus graphics do not connect page to page, so page labels are used
U12
8085A
C12
100nf, 50V
+5V
READY
HOLD
INTR
RST_5.5
RST_6.5
RST_7.5
TRAP
RES_IN
X1
X2
SID
SOD
VCC
VSS
AD0
AD1
AD2
AD3
AD4
AD5
AD6
AD7
A8
A9
A10
A11
A12
A13
A14
A15
ALE
S0
S1
IO/M
RD
WR
HLDA
INTA
RES_OUT
CLK
RDY
HOLD
INTR
RST5.5
RST6.5
RST7.5
TRAP
/RESET
cpuA8
cpuA9
cpuA10
cpuA11
cpuA12
cpuA13
cpuA14
cpuA15
OLD_PSYNC
install jumpers to use Y2
Y1
Crystal
C26
20pF, 25V
note: use C26 only when Y2<=4MHz
SID_SOD1
GND
C5
100nf, 50V
R7
10K
R8
4.7K
Y3
EN
OUT
2MHZ
FP
Front Panel
U13B
74HCT04
U13A
74HCT04
U13C
74HCT04
U3B
74HCT74
/ALE
/IPOC
R9
1K
EXT_CLR*
1N4148
CR1
U18A
74HCT367
/OE1
CLOCK*
POC*
Phase2
Phase1
U18B
74HCT367
/OE2
U2D
74HCT02
sOUT
Note: IMSAI MPU-B had PCB jumper trace on bottom side Copper to Ground
J3
MWRT
J11
M1
PSYNC
T2LONGER
T2_LONGER
U17A
74HCT367
pDBIN
pSYNC
pWAIT*
pWR*
/RD
/WR
LOCAL_PSYNC
R4
470
R5
470
R6
470
U4A
74HCT00
/INTA
U17B
74HCT367
pHLDA
R1
470
R2
470
C6
100nf, 50V
Y2
U4B
74HCT00
NOTE: 6/16/2026 special Oscillator Foot Prints allow either DIP8 or DIP14 to be used for Y2 and Y3
NOTE: Y2 upper limits are 12MHz for Intel 8085A-1 (6MHz); 16MHz for Tundra CA80C85B (8MHz); memory, IO cards and back plane might limit highest speed to less
R3
1K
U4C
74HCT00
CDSB*
/CCDSBL
8085 CPU & Support Signals
Sheet: /8085CPU/
File: CPU8085_to_S100.kicad_sch
Title:
Size: A4
Date: 17 July 2026, greglori_mi@wowway.com
Rev: rev 1.04
KiCad E.D.A. 10.0.4
Id: 5/12



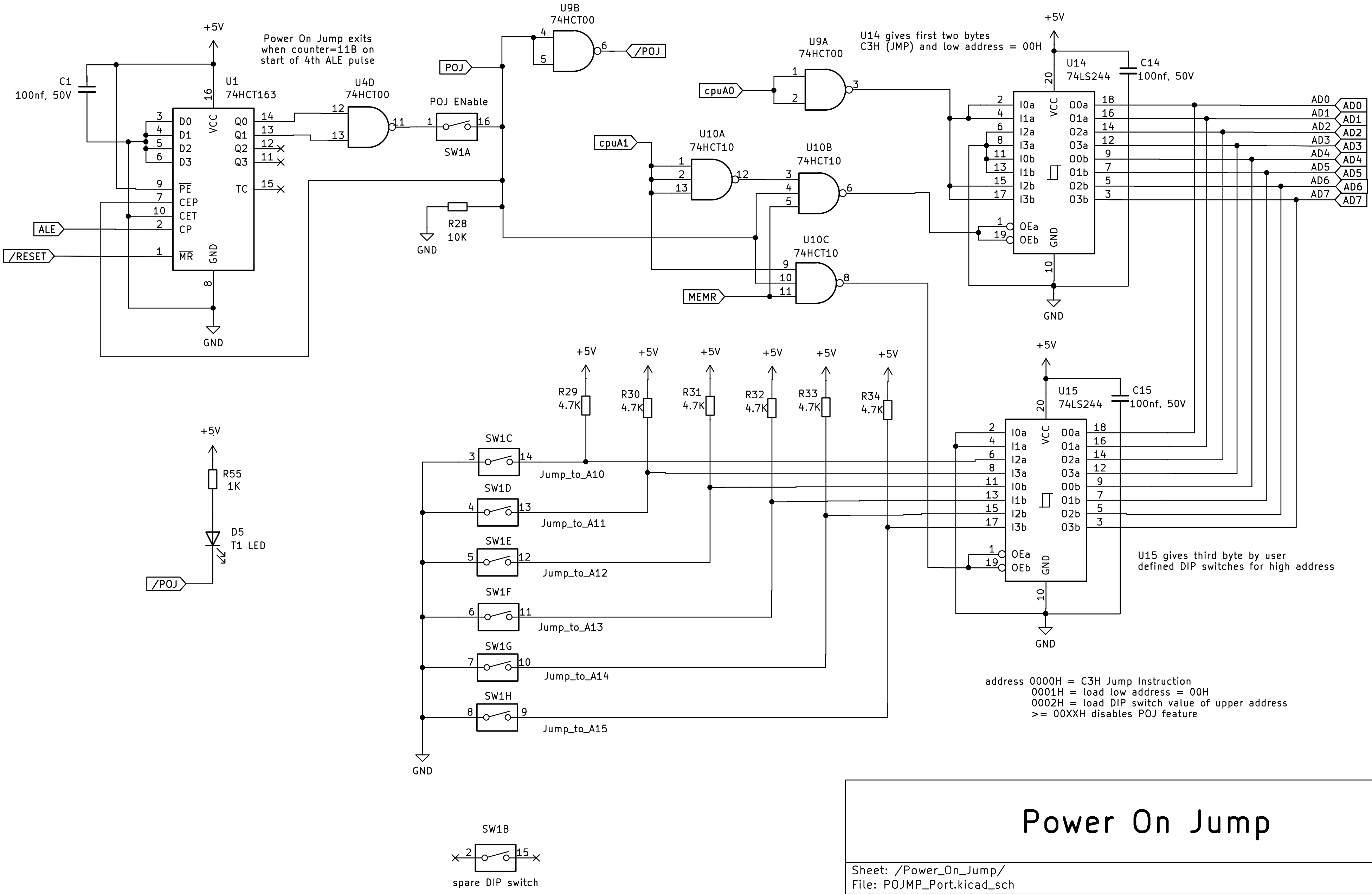
Board Buses to/from S-100

Sheet: /S-100_Address_and_Data_Bus (page 6)/
File: AandDBus.kicad_sch

Title:

Size: A4 | Date: 17 July 2026, greglori_mi@wowway.com | **Rev:** rev 1.04

KiCad E.D.A. 10.0.4 | Id: 6/12



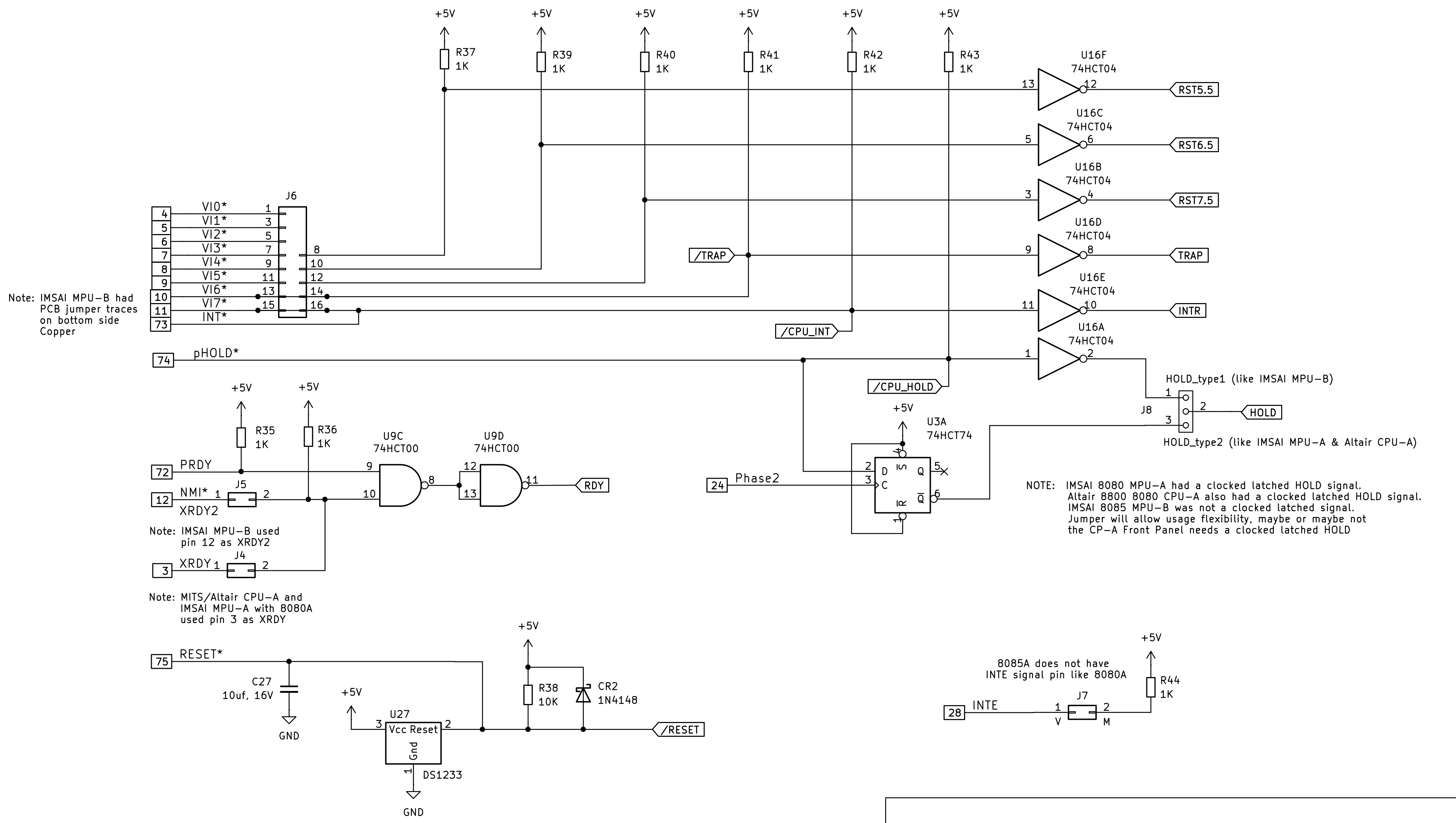
Power On Jump

Sheet: /Power_On_Jump/
File: POJMP_Port.kicad_sch

Title:

Size: A4 | Date: 17 July 2026, greglori_mi@wowway.com | **Rev:** rev 1.04

KiCad E.D.A. 10.0.4 | Id: 7/12



NOTE: IMSAI 8080 MPU-A had a clocked latched HOLD signal.
Altair 8800 8080 CPU-A also had a clocked latched HOLD signal.
IMSAI 8085 MPU-B was not a clocked latched signal.
Jumper will allow usage flexibility, maybe or maybe not
the CP-A Front Panel needs a clocked latched HOLD

NOTE: 8080A and early EPROMs needed -5V, +12V and 5V to power up "in spec" together.
8085A needs simpler +5V only, but still needs inhibit <4.5V.
IMSAI and Altair CPU cards would not start correctly with a slow rising +5Vcc.
DS1233 output is logic 0 when 5Vcc <4.5V (for any +5V ramp up speed)
plus it gives an extra 350mS duration before releasing reset.

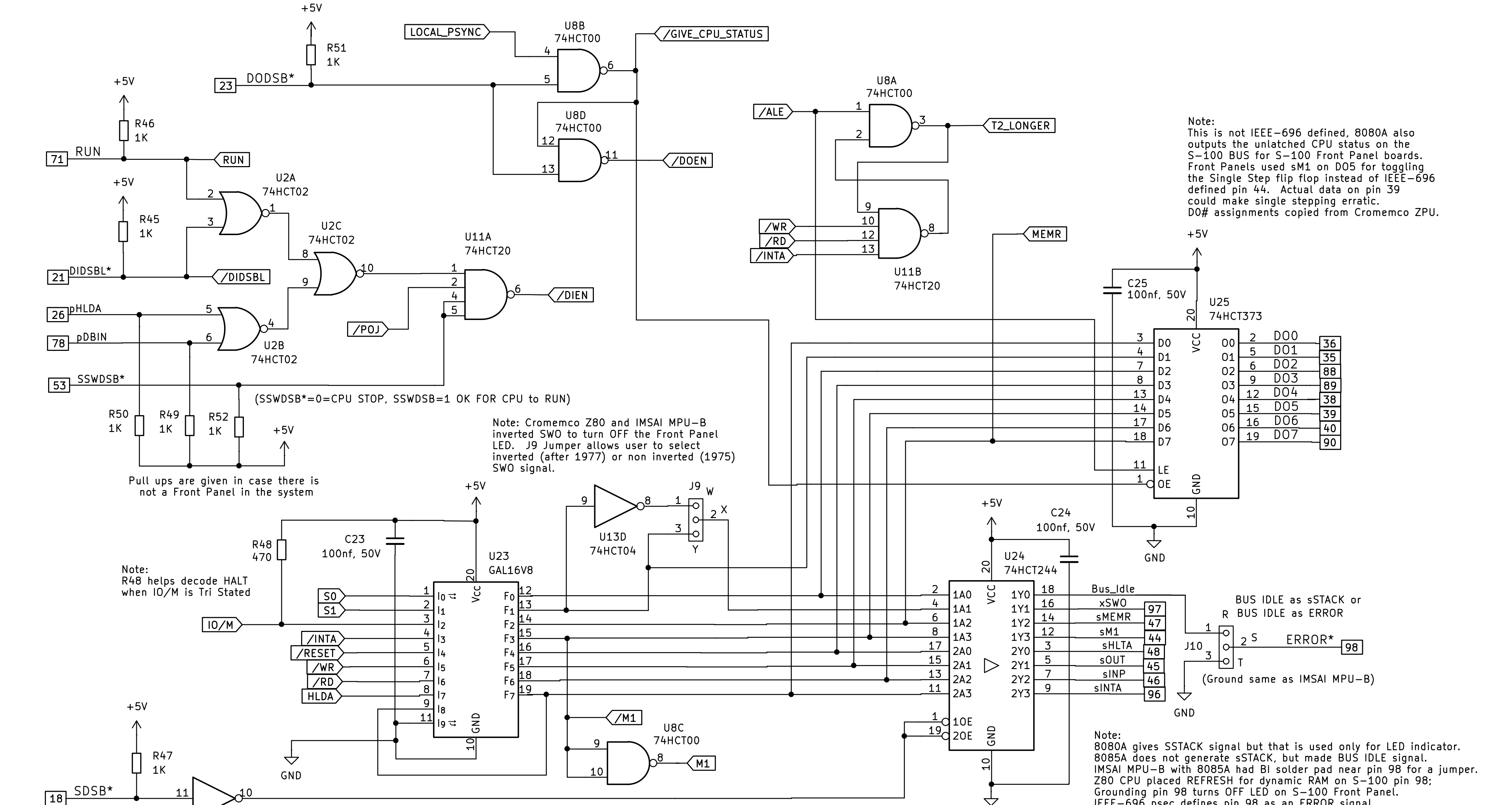
Interrupt Vectors

Sheet: /Interrupt Vector Signals/
File: Interrupts.kicad_sch

Title:

Size: A4 | Date: 18 July 2026, greglori_mi@wowway.com | **Rev:** rev 1.04

KiCad E.D.A. 10.0.4 | Id: 8/12



STATUS WORD CHART

TYPE OF MACHINE CYCLE

DATA BUS BIT	STATUS INFORMATION	① INSTRUCTION FETCH	② MEMORY READ	③ MEMORY WRITE	④ STACK READ	⑤ STACK WRITE	⑥ INPUT READ	⑦ OUTPUT WRITE	⑧ INTERRUPT ACKNOWLEDGE	⑨ HALT ACKNOWLEDGE	⑩ INTERRUPT ACKNOWLEDGE WHILE HALT
D_0	INTA	0	0	0	0	0	0	0	1	0	1
D_1	$\overline{WO}$	1	1	0	1	0	1	0	1	1	1
D_2	STACK	0	0	0	1	1	0	0	0	0	0
D_3	HLTA	0	0	0	0	0	0	0	0	1	1
D_4	OUT	0	0	0	0	0	0	1	0	0	0
D_5	M_1	1	0	0	0	0	0	0	1	0	1
D_6	INP	0	0	0	0	0	1	0	0	0	0
D_7	MEMR	1	1	0	1	0	0	0	0	1	0

Ⓝ STATUS WORD

Table 2-1. 8080 Status Bit Definitions

CPU State Decoding

Sheet: /CPU_States/
File: CPUstates.kicad_sch

Title:

Size: A4 | Date: 18 July 2026, greglori_mi@wowway.com | **Rev:** rev 1.04

KiCad E.D.A. 10.0.4 | Id: 9/12

ASSEMBLY INSTRUCTIONS:

1) solder lowest height parts first... resistors, axial diodes

2) then solder DIP sockets, the ceramic capacitors, LED

3) and then the 2 and 3 pin SIP headers

4) solder tallest parts radial electrolytic capacitors and the VR1 & its optional heat sink last

5) use black electrical tape to cover gold electrical edge card fingers and any front to back PCB holes to prevent PCB cleaner seeping under front side parts

6) hold PCB vertically and apply about 3 oz, or 1 even coating, over the rear of the PCB of the ZEP Critrus cleaner

7) wait about 1 minute

8) hold PCB vertically, apply about 3 oz to 4 oz or 1 even washing coat of Kwik Clean Flux Cleaner

9) wait about 1 minute

10) rinse with distilled water

11) shake water off the PCB or use a hair dryer on "Cold" high speed setting to blow dry the PCB

12) power up the panel without ICs to check the regulator output voltage (5V)

13) without power applied to the panel, install ICs logic, CPU, etc. (Do not "Hot Plug" parts or do not install parts with power active.)

14) looking down onto the top of the PCB, look for any outward bent pins that did not go into their socket receivers, remove, fix, un-bend and reinstall parts if necessary

15) look from the PCB S-100 edge under all ICs, check for any unseen bent, folded under pins, remove, fix, un-bend and reinstall parts if necessary

16) set DIP switches and jumper blocks to desired configuration

17) power up the CPU-85 and go.... use as needed with your S-100 application(s)

ASSEMBLY INSTRUCTIONS		
Sheet: /ASSEMBLY INSTRUCTIONS/ File: Assembly_sch.kicad_sch		
Title:		
Size: A4	Date: 14 July 2026, greglori_mi@wowway.com	**Rev:** rev 1.04
KiCad E.D.A. 10.0.4		Id: 10/12

CONFIGURATION OPTIONS:

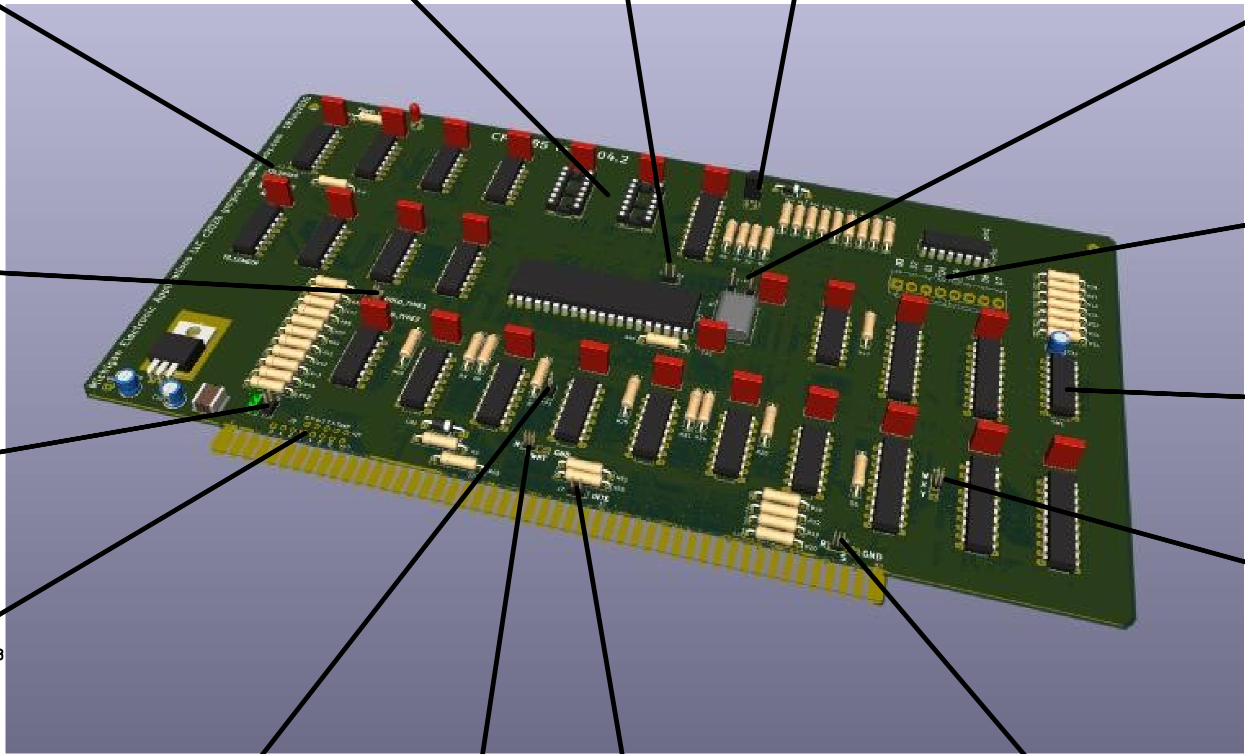
DS1233 +5V Monitor, Low Voltage Inhibit (LVI)
note: almost all CPUs from 1970s lack LVI, MicroChip and Motorola made LVI a standard feature in late 1980s
* do not populate if 1970s performance is desired (not recommended)
(recommend installing to avoid start up and shut down glitching issues on floppy drives)

J11 PSYNC source
* use ALE address latch like IMSAI MPU-B
* use M1 from CPU State Decoder (like Cromemco Z80 board)
* use simulated signal lengthen to last till end of T2 cycle like a 8080A on IMSAI and MITS/ALTAIR boards

install Y2 = maximum allowed by users system

Solder pads for
SID and SOD of 8085

Y1 / Y2 Jumpers
* no juempers, allows use of Y1 quartz crystal per Intel 8085 data sheet page 6 figure 5
* or PDIP oscillator Y2 as CPU clock source (my preference is for Y2 as DIP oscillator)



Ribbon Cable Interfaces
* 16 pin PDIP for IMSAI Front Panel
* 8 pin Molex for MITS/Altair Front Panel

J8 HOLD - RDY circuit source
* Type1, no latch like IMSAI MPU-B
* Type2, latched like MITS/Altair CPU-A

see Intel 8085 data sheet page 6 Fig. 6

Power On Jump, DIP switches
* switch 1 = ON enables power on Reset Jump
* switch 1 = OFF uses CPU default 0000H address
* switch 2, not used
* switched 3 to 8, Address A10 to A15, ON=logic 0 (1K increments from 0100H to FC00H)

XRDY and XRDY2 sources
* J4 jumper use pin 3 like MITS/Altair
* J5 jumper use pin 12 like IMSAI MPU-B
* can select both as a "wired OR"

J9 xSWO source for S-100 pin 97
* IMSAI MPU-B and Cromemco Inverted xSWO (W-X jumper) turns Front Panel LED default OFF default (a 1977 corrective action)
* or IMSAI MPU-A MITS/Altair xSWO signal (X-Y jumper) as implemented in 1975 but will leave Front Panel LED ON by default (an IMSAI MITS/Altair oops)

J6 Interrupts 5.5, 6.5 and 7.5
note: 8085 contains 3 vectors of a 8214
* jumpers select 3 interrupt sources 1 thru 8

J7 INTE S-100 pin 28 (V-M jumper)
note: 8085 does not have an INTE output
* Open Circuit or
* pull up voltasge to +5V (like IMSAI MPU-B)

ERROR or SSTACK source for S-100 pin 98
note: 8085 does not make the SSTACK output
* Ground to disable Front Panel LED (S-T jumper) like IMSAI MPU-B
* or decoded BUS IDLE CPU State (R-S jumper)
* MIYS/Altair CPU-A and IMSAI MPU-A called this SSTACK
* IEEE-696 defines 98 as bus ERROR signal

FP1, allow CPU to drive MRWT S-100 bus signal pin 68
* Open Circuit let Front Panel drive MWRT
* if no Front Panel, then install jumper to let CPU drive MWRT this is MWRT Enable for J3

J3 MWRT Source S-100 pin 68
* CPU driven signal (N-P jumper) also install FP1 for MWRT Enable
* Ground (P-O jumper)

PCB OPTIONS		
Sheet: /CONFIGURATION OPTIONS/ File: Options_sch.kicad_sch		
Title:		
Size: A4	Date: 17 July 2026, greglori_mi@wowway.com	**Rev:** rev 1.04
KiCad E.D.A. 10.0.4		Id: 11/12

JUMPER OPTIONS USED/ VALIDATION SET UP:

1) Set Power On Jump to go to F800H (for IMSAI PCS Monitor program), OFF for IMSAI and MITS/ALTAIR Front Panels
2) Y1 and C26 not installed, jumpers installed to use Y2=8MHz oscillator until I determine highest feasible MHz
3) FP jumper installed to allow CPU to control MWRT, MWRT jumper N-P
4) J4 and J5, I installed both jumper blocks to support either MITS/Altair and IMSAI
5) J6 no interrupts or jumpers used (user's application specific)
6) J7 jumper to have pull up to 5V for INTE (same as IMSAI MPU-B)
7) J8 jumper HOLD_TYPE1 closed (same as the IMSAI MPU-B)
8) J9 jumper X-Y for not inverting (U24 output is programmed to make the inverted /sWO for 1977)
9) J10 jumpered S-T to Ground (original IMSAI MPU-B setting)

VALIDATION PLANS and RESULTS (RFU, living document):

These procedures are for me, the PCB designer, to validate and verify.
These are FYI for the general user to know the actions I did to check this PCB design (and to be a note to myself to help me know what I did in the future).

1) (PASS) KiCAD 10.0.4 Design Rule Checking (DRC) tool
   ** Drc report for CPU-85_r2c.kicad_pcb **
   ** Created on 2026-07-06T12:05:31 **
   ** Report includes: Errors, Warnings **
   ** Found 0 DRC violations **
   ** Found 0 unconnected pads **
   ** Found 0 Footprint errors **

2) new card must insert and egress with reasonable force from 1 hand into and out of S-100 female receiving connector on S-100 back plane.

3) new card must slide freely between existing IMSAI S-100 back plane card cage guides.

4) take 2 blank or production fresh panels, install each into 2 different S-100 female connector manufacturer brands 5 times,
   visually inspect the female connector's witness marks on the edge cnnectors for consistant lcoation across all edge pins
   (I dont want to wear out my IMSAI back plane's connectors, IEEE spec has a 50 insertion cycle minimum)
   black original IMSAI back plane connector:
   green Mullen Extended Board:

5) solder on all parts & sockets after receiving new panels, and install all ICs per Assembly Instructions

6) install different Y2 oscillators to find the upper speed limitation of my PCS80/30.
   Look for abnormal text on video display (this operation might take 5 minutes per run, enough time for parts top warm up and shift performance if theres a timing issue.
   2.00 MHz (for CPU speed = 1MHz or XTAL/2)
   8.00 MHz
   10.00 MHz
   11.00 MHz
   12.00 MHz
   14.00 MHz
   15.00 MHz
   16.00 MHz
   make a PIC12C675 toggle outputs at 0.5Hz into a 80C85B, see if 80C85B can run that slow

7) Keep the computer and CPU card powered for 30 minutes with Y2 = highest speed, and take a thermal image to determine if any components are above
   45 degress Celsius or more than 20 degrees above ambient. (higher temperatures indicate abnormal or unexpected loading)

8) change jumper from HOLD Type1 to HOLD Type2, and observe if there is an observable difference in operation

9) Plug in Don Caprio's repro version of IMSAI Front Panel and observe what feaatures operate and which do not operate.
   RESET (Power On Jump Active LED should flash "on" at power up and stay "off")
   STOP
   RUN
   SINGLE STEP
   EXAMINE

10) ask S100 Vintage Computer Group member with an Altair to try out and return test panel since I don't have a MITS/Altair8800.

VALIDATION PLAN & RESULTS

Sheet: /VALIDATION PLANS/
File: Validation_sch.kicad_sch

Title:		
Size: A4	Date: 17 July 2026, greglori_mi@wowway.com	**Rev:** rev 1.04
KiCad E.D.A. 10.0.4		Id: 12/12